Khyati Khaitan

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Education

Vellore Institute of Technology August, 2020 - July 2022
Master of Computer Applications *CGPA: 9.13*

West Bengal University of Technology August, 2016 - July, 2019
Bachelor of Computer Applications *CGPA: 8.2*

Experience

AI Product Engineer December 2025 – Present
Voiceplug Inc. *Remote*

- Working on production-grade AI voice ordering features for a multi-tenant QSR platform across multiple live restaurant deployments, improving conversational reliability and customer ordering experience.
- Improved live ordering success rates from 25–30% to 60%+ by resolving entity ambiguity, menu-understanding failures, ASR inconsistencies, and order-validation edge cases across POS-integrated systems.
- Optimized scalable real-time ordering workflows using FastAPI, Socket.IO, Redis, MongoDB, and Jinja2-based prompt templating while enhancing intent recognition, fallback handling, and conversational flow reliability.
- Collaborated with product, QA, and support teams to analyze 500+ real customer interactions and translate insights into production improvements, reducing ordering friction and improving deployment readiness across live QSR environments.

Software Engineer July 2022 – November 2025
Bank of America *Gandhinagar,GJ*

- Developed a real-time backend chatbot application using the RASA framework for team member access retrieval, reducing manual checks and improving efficiency by 100%. This solution received recognition from senior management for saving significant time and effort.
- Developed a database parameter comparison utility, optimizing analysis and eliminating mismatches across environments by 100%.
- Conducted a detailed regression analysis, preparing baseline and post-baseline reports to detect any anomalies, and performed thorough ORA-error analysis post-patch, confirming successful patch application and system integrity.
- Led a high-impact domain decommissioning project. Executed a recursive dependency analysis on 140+ impacted files, successfully refactoring and updating 50+ outdated Python scripts to resolve critical system issues and reducing legacy infrastructure maintenance costs by 20%.

Software Intern August 2021-April 2022
Oracle Cerner *Bengaluru,KA*

- Part of the Data Migration team where got entrusted with the job of extracting, transforming and then exporting the confidential clinical data from the company's Millennium application to the client.
- Responsible for writing HL-7 , CCL scripts and modified them according to the data which needed to be migrated.
- Leveraged Unix shell scripting and tools like awk,grep, and sed to extract, process and store data from Millenium on Unix Systems

Personal Projects

Bengali-to-Hindi Text Refiner | *Python, PyTesseract, M2M100, Flan-T5* 2025

- Designed a pipeline to extract handwritten Bengali text from images and translate it into fluent, readable Hindi.
- Used PyTesseract for OCR, M2M100 for multilingual translation, and Flan-T5 to enhance fluency and readability.

Finance News Research Tool |*Python,Streamlit,LangChain,FAISS,HuggingFace,Groq* 2025

- Built a RAG-based finance news Q A tool to extract article text from URLs and split it into overlapping chunks.
- Generated Hugging Face MiniLM embeddings and indexed chunks in FAISS for persistent semantic search.
- Integrated Groq via LangChain for context-grounded answers and displayed retrieved source URLs in Streamlit.

Achievements and Leaderships

Gold Award: For single-handedly executing multiple high-impact projects in a short span of 1 year without any production failures or downtime.
Silver Award: For demonstrating adaptability and commitment by delivering various complex modules on time.
Recognition: Recognized by the upper management for efficiently managing a critical weekend data purge, ensuring smooth execution and zero downtime
DSA Proficiency: Solved 570+ Data Structures and Algorithms (DSA) problems on LeetCode, enhancing problem-solving skills across various topics such as dynamic programming, graph theory, and optimization.
Student Lead: Anti-Ragging Cell, MAKAUT

Technical Skills

Languages: Python, Java, SQL, Shell Scripting
AI GenAI & NLP: RAG, LangChain, LangGraph, Google Gemini, Groq, Hugging Face, FAISS, Prompt Engineering, Conversational AI, Voice AI
Frameworks & APIs: FastAPI, Socket.IO, Streamlit, Django, Django REST Framework, Rasa
Databases & Caching: MongoDB, Redis
Tools & Platforms: Git, Bitbucket, Jira, Linux/Unix, VS Code, PyCharm, Google Colab
Concepts: Semantic Search, Vector Embeddings, Text Chunking, Intent Recognition, Entity Extraction, ASR Error Handling, Fallback Handling, Prompt Templating, POS Integrations, Real-Time Voice AI Systems